



Superior reduction quality without improved outcomes: open versus closed reduction for displaced femoral neck fractures in adults under 60 years

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Abstract

Introduction The optimal reduction strategy for displaced femoral neck fractures (DFNFs) in non-elderly adults remains controversial. Although open reduction may improve radiographic alignment, its impact on outcomes remains unclear.

Materials and methods Adults aged < 60 years with DFNFs who were treated with internal fixation between 2017 and 2023 were retrospectively reviewed. Patients were categorized into closed and open reduction groups. Reduction quality was assessed using the Garden alignment index and the Haidukewych reduction criteria. Multivariate analyses were performed for the overall cohort and patients with Pauwels type III fractures. Outcomes included non-union, osteonecrosis, femoral neck shortening, and major reoperations.

Results In total, 98 patients were included, among whom 83 and 15 underwent closed and open reduction, respectively. Although open reduction achieved a higher rate of excellent reduction (53.3% vs. 16.9%, $p=0.012$), it was not associated with lower rates of non-union or osteonecrosis. In the overall cohort, age was associated with non-union (odds ratio [OR], 1.09; $p=0.024$), whereas poorer reduction quality was associated with major reoperation (fair: OR 7.03, $p=0.012$; poor: OR 47.6, $p=0.007$). In patients with Pauwels type III fractures, poorer reduction quality predicted conversion to arthroplasty (fair: OR 9.84, $p=0.014$), and age was associated with non-union and osteonecrosis. Femoral neck shortening of > 10 mm at 3 months was associated with age and open reduction, whereas at 12 months, age and intraoperative blood loss remained significant predictors.

Conclusions Open reduction achieved a higher rate of excellent reduction but did not reduce complication rates compared with closed reduction. Fracture severity, patient age, and reduction quality remained the principal determinants of outcome, supporting a fracture morphology-guided fixation strategy and realistic patient counseling.

Keywords Femoral neck fracture · Reduction quality · Reduction strategy · Femoral neck shortening · Fixation failure

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Introduction

Displaced femoral neck fractures (DFNFs) in individuals younger than 60 years are uncommon, with an estimated annual incidence of 4–72 per 100,000 people [1, 2]. Although mortality is lower than that in older patients, these fractures typically result from higher-energy trauma and are often more severe [3, 4]. Moreover, disruption of the femoral head vascular supply is thought to contribute to an increased risk of non-union and osteonecrosis in this population [5]. Reported rates of treatment failure, including fixation failure, non-union, symptomatic malunion, and osteonecrosis, vary substantially by fracture pattern, reduction quality, and fixation strategy, ranging from 5 to 90%

[6–9]. Approximately 15% of young adults with DFNFs eventually require conversion to total hip arthroplasty, usually within two years [9]. However, the age threshold defining “young” or “non-elderly” patients remains inconsistent across the literature. Although many studies have used 50 years as the cutoff, contemporary treatment strategies frequently extend head-preserving fixation to physiologically active patients approaching 60 years of age [10, 11].

Joint preservation is the goal of physiologically young patients; however, the optimal surgical approach remains debatable. Reduction quality is the most critical surgeon-controlled determinant of outcomes and is strongly associated with uneventful union, whereas inadequate reduction increases the risk of treatment failure by more than five-fold [6, 12, 13]. Although open reduction is often advocated to achieve anatomical reduction, pooled analyses have reported similar rates of acceptable reduction between open and closed techniques, with open reduction associated with higher rates of reoperation and complications [6, 12–17].

In contrast, closed reduction relies on fluoroscopic assessment of alignment and displacement. Despite its presumed advantage in preserving femoral head vascularity, closed reduction has poor to moderate inter- and intra-observer reliability, limiting the reproducibility of reduction quality [18]. Postoperative femoral neck collapse, characterized by varus deformity or shortening, may result from inadequate reduction, unstable fracture patterns, injury mechanisms, or implant selection and is associated with poorer functional outcomes [19].

The purpose of this study was to compare the reduction quality and postoperative radiographic outcomes between closed and open reduction in adults younger than 60 years with DFNFs. The primary outcome was the reduction in quality, assessed using the established criteria. Secondary outcomes included radiographic stability, femoral neck collapse and shortening, as well as fixation failure, including non-union, osteonecrosis, and major reoperations.

Methods and materials

Patient enrolment and data collection

Medical and radiographic records of patients younger than 60 years with DFNFs who underwent osteosynthesis between January 2017 and June 2023 were retrospectively reviewed using the Institutional Trauma Registry. Displacement was defined as a Garden type III or IV fracture [20]. Patients with bilateral fractures, pathological fractures, or prior hip surgery were excluded.

Eligible patients were categorized into closed and open reduction groups based on the reduction technique applied.

Demographic characteristics, injury-related variables, and operative details were systematically extracted from electronic medical records and operative reports. Attending orthopedic trauma surgeons or fellowship-trained surgeons performed all the surgeries. In cases where orthopedic residents participated as primary operators, the procedures were performed under the direct supervision of the attending orthopedic trauma surgeons. This study was approved by our Institutional Review Board (IRB No. 202600177B0).

Surgical strategy

Osteosynthesis was generally performed within 24 h of DFNF confirmation, unless contraindicated. The choice of closed or open reduction was determined by the treating surgeon’s clinical judgment based on fracture morphology, intraoperative reducibility, and overall patient condition.

For closed reduction, patients were positioned supine on a traction table, and reduction was achieved under fluoroscopic guidance (Ziehm Solo; Ziehm Imaging GmbH, Nuremberg, Germany), with provisional femoral head control using a percutaneously inserted Kirschner wire used as a joystick (Fig. 1). Reduction quality was assessed on anteroposterior and lateral femoral neck views, and definitive fixation was performed after satisfactory alignment was confirmed. Fixation constructs included three or four percutaneous cannulated screws (CS, Syntec Technology Co., Hsinchu, Taiwan), a sliding hip screw (SHS) with an anti-rotation screw (Syntec Technology Co.), the Femoral Neck System (FNS, DePuy Synthes, Paoli, PA, USA), or the INTERTAN™ cephalomedullary nail (Smith & Nephew, London, UK).

For open reduction, the patients were positioned supine on a radiolucent table (Modular Table System, Mizuho OSI, Union City, CA, USA), and reduction was performed using the Smith–Petersen approach [21]. These procedures were primarily performed by senior orthopedic trauma surgeons with experience in femoral neck preservation surgery. After exposure of the anterior hip capsule, a Z-shaped capsulotomy was performed to directly visualize the fracture site and facilitate anatomical reduction. Care was taken to avoid excessive posterior-superior capsular stripping and to preserve the retinacular soft tissues supplying the femoral head. Fracture fixation was then performed under combined direct visualization and fluoroscopic guidance. In most cases, an additional medial buttress plate (MBP) using a one-third semitubular plate (Syntec Technology Co.) was added to enhance stability (Fig. 2). Definitive fixation implants were otherwise identical to those used for closed reduction.

Fig. 1 Intraoperative fluoroscopic views of a displaced femoral neck fracture demonstrating closed reduction on a traction table. Anteroposterior (a) and lateral (b) views are shown. A percutaneously inserted Kirschner wire (arrow) was used as a joystick to achieve and maintain provisional femoral head control under fluoroscopic guidance. Postoperative radiographs (c, d) show definitive fracture fixation with multiple cannulated screws to ensure construct stability

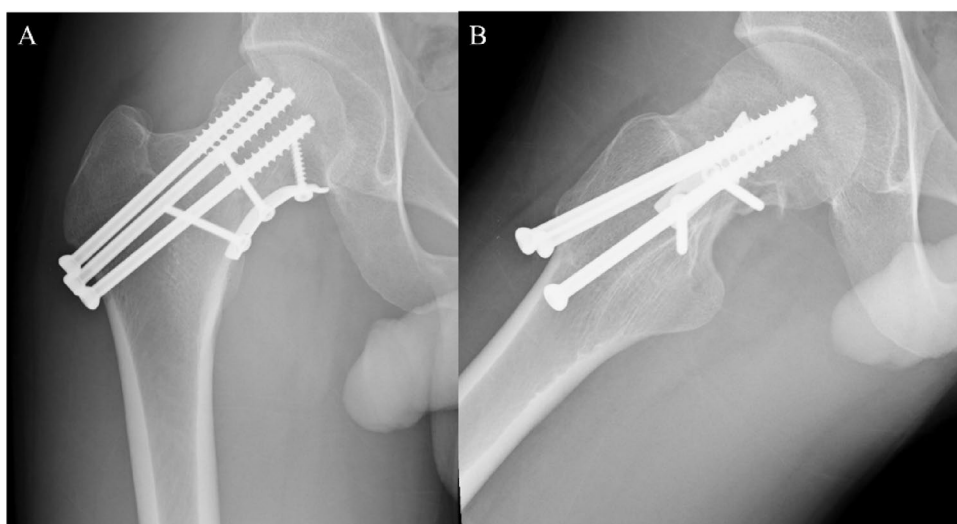
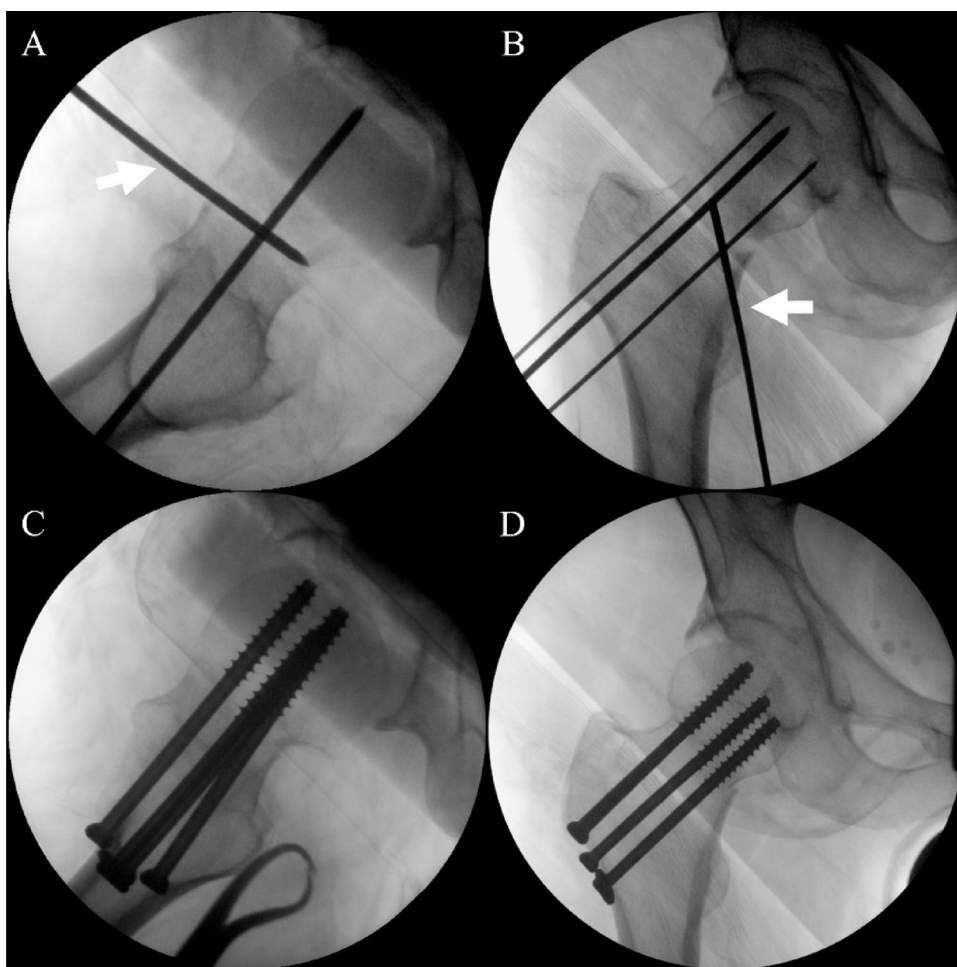


Fig. 2 Postoperative radiographs. Anteroposterior (a) and lateral (b) views show fixation with multiple cannulated screws and a one-third semi-tubular plate used as a medial buttress to provide additional construct stability



Postoperative image follow-up and rehabilitation protocol

Radiographic follow-up, including anteroposterior and lateral views of the hip, was routinely performed on

postoperative day 1, as well as at 4 weeks, 3, 6, and 12 months. Passive and active range-of-motion exercises of the affected hip were initiated immediately after surgery without restriction. Non-weight-bearing ambulation with crutches or a walker was prescribed for the first 4 weeks, followed by gradual progression to partial and then full weight-bearing

by 3 months, unless contraindicated. For patients unable to use crutches or a walker, such as those with concomitant upper-extremity injuries, ambulation was assisted using a wheelchair.

Applied classifications

Femoral neck fractures were classified using the Garden and Pauwels systems [20]. The Garden classification categorizes fractures into types I to IV based on fracture completeness and displacement [20]. Pauwels' classification was determined according to the inclination of the fracture line relative to the horizontal plane: type I, $< 30^\circ$; type II, 30° – 50° ; and type III, $> 50^\circ$ [20].

Reduction quality was assessed using two established indices. The Garden alignment index (GAI) was measured on postoperative anteroposterior and lateral radiographs to evaluate the alignment of femoral head trabeculae relative to the femoral shaft [22]. Based on the GAI, the reduction quality was categorized as anatomical, acceptable (155° – 180° in both views), borderline ($< 155^\circ$ or $> 180^\circ$ in either view), or unacceptable ($< 150^\circ$ or $> 185^\circ$ in the anteroposterior view).

Reduction quality was further evaluated using the Haidukewych reduction criteria (HRC) [12]. The HRC provides a quantitative assessment of femoral neck fracture reduction based on the maximum residual displacement or angulation measured on postoperative anteroposterior and lateral radiographs. Reduction was graded as excellent (< 2 mm displacement or $< 5^\circ$ angulation in any plane), good (2–5 mm displacement or 5 – 10° angulation), fair (5–10 mm displacement and/or 10 – 20° angulation), or poor (> 10 mm displacement, $> 20^\circ$ angulation, or residual varus alignment).

Femoral neck shortening was assessed using radiographic measurement methods described by Zlowodzki et al. and Polat et al. [19, 23]. Shortening was determined by comparing serial anteroposterior radiographs of the affected hip, and clinically relevant shortening was defined as a decrease of ≥ 10 mm. Posteromedial cortical comminution was defined as the absence of a continuous opposing posterior and medial femoral neck cortex on postoperative anteroposterior or lateral radiographs [16].

Interobserver reliability was assessed by two orthopedic trauma surgeons, with discrepancies resolved by a senior surgeon.

Definition of complications

Major complications included non-union, osteonecrosis, or major reoperations. Non-union was defined as the absence of radiographic healing at 6 months and/or loss of fixation integrity [24]. Osteonecrosis was diagnosed and staged

according to the Association Research Circulation Osseous classification [25], with stage III or higher considered a failure. Major reoperation included conversion to total hip arthroplasty or hemiarthroplasty, muscle pedicle bone grafting, or early fixation revision. Elective implant removal after fracture union was not considered a complication.

Statistical analysis

Data are presented as the mean $\pm$ standard deviation. All statistical analyses were performed using SPSS version 22 software (IBM Corp., Armonk, NY, USA). Continuous and categorical variables were compared using independent t-tests and the chi-square test or Fisher's exact test, respectively. Femoral neck shortening was measured using two-way ANOVA, and neck-shaft angle (NSA) was analyzed using the Mann–Whitney U test and Friedman test for inter- and intragroup comparisons at each time point. Intraobserver reliability was assessed using intraclass correlation coefficients (ICC) (95% confidence interval [CI]) and Fleiss-Kappa (κ). Stepwise multivariate logistic regression was used to identify predictors of adverse outcomes. Statistical significance was set at $p < 0.05$.

Results

Patient characteristics

During the study period, 98 patients met the inclusion criteria (closed, $n = 83$; open, $n = 15$) (Table 1). Patients in the open reduction group were younger (39.6 ± 12.2 vs. 46.3 ± 10.8 years, $p = 0.03$). The open reduction group had a longer operative time (195.9 ± 47.0 vs. 130.2 ± 78.9 min, $p = 0.002$) and greater intraoperative blood loss (296.7 ± 262.2 vs. 92.6 ± 199.4 mL, $p = 0.001$) than the closed reduction group. In the open reduction group, the primary fixation devices included cannulated screws in 12 patients, SHS in 1 patient, and FNS in 2 patients. Fourteen of the 15 patients additionally received medial buttress plate augmentation, whereas 1 patient underwent open reduction with cannulated screw fixation alone.

Complications and hip survival

During follow-up, osteonecrosis and non-union occurred in 17 (17.3%) and 23 (23.5%) patients, respectively (Table 1). Non-union was more frequent in the open reduction group (46.7% vs. 19.3%, $p = 0.04$), whereas the incidence of osteonecrosis did not differ between the groups.

Overall, 30 patients (30.1%) underwent major reoperation. Of these, 27 (27.6%) ultimately required conversion to

Table 1 Demographic characteristics and perioperative data of the 98 patients

	Total (n=98)	CR group (n=83)	OR group (n=15)	p value
Age	45.3±11.3	46.3±10.8	39.6±12.2	0.03*
Sex				0.77
Male	33	29	4	
Female	65	54	11	
BMI	23.4±3.6	23.3±3.5	23.7±4.2	
Smoking				0.66
Yes	28	23	5	
No	70	60	10	
Alcohol				0.91
Yes	14	12	2	
No	84	71	13	
Affected side				0.01*
Left	54	41	13	
Right	44	42	2	
Injury mechanism (n)				0.08
Motor vehicle accident	45	34	11	
Simple fall	38	36	2	
Fall from height	14	12	2	
Crush injury	1	1	0	
Admission duration (day)	7.8±9.8	6.4±5.6	15.7±20.0	0.09
Follow up (month)	23.8±17.7	22.9±17.3	28.7±19.7	0.24
Injury to operation (hour)	37.8±62.8	35.9±63.3	48.0±61.2	0.49
Operation time (minute)	140.3±78.4	130.2±78.9	195.9±47.0	0.002*
Estimated blood loss (mL)	123.8±221.3	92.6±199.4	296.7±262.2	0.001*
Garden classification				0.19
III	93	78	15	
IV	5	5	0	
Implant				0.304
Cannulated screws	66	65	12	
HCS with screw(s)	11	11	1	
FNS	3	3	2	
IMN	4	4	0	
Medial buttress plate augmentation				
Yes		0	14	
No		83	0	
Hip survival (month)	20.2±18.6	19.8±16.6	21.7±26.2	0.83
Complication				
Osteonecrosis	17 (17.3%)	13 (15.7%)	4 (26.7%)	0.32

Table 1 (continued)

	Total (n=98)	CR group (n=83)	OR group (n=15)	p value
Non-union	23 (23.5%)	16 (19.3%)	7 (46.7%)	0.04*
Major reoperation†	30 (30.1%)	23 (27.7%)	7 (46.7%)	0.22

HCS hip compression screw, FNS Femoral neck system, IMN intra-medullary nail, BMI body mass index

* Statistically significant

†Major reoperation included conversion to arthroplasty, muscle pedicle bone grafting, and early revision of internal fixation

hip arthroplasty. The remaining 3 (2.5%) patients were successfully treated with head-preserving salvage procedures. Major reoperation rates (46.7% vs. 27.7%; $p=0.22$) and mean hip survival time did not differ significantly between the groups ($p=0.83$). Two patients underwent early revision of internal fixation; salvage of fixation was achieved in one case, whereas the other later required arthroplasty because of non-union. Three patients underwent muscle pedicle bone grafting for non-union; successful salvage of fixation was achieved in one case, whereas the remaining two later required arthroplasty because of persistent non-union. One patient underwent core decompression with autologous bone grafting for early-stage osteonecrosis and was successfully salvaged.

Radiographic outcome

Postoperative radiographic outcomes are summarized in Table 2. Pauwels type III fractures were more prevalent in the open reduction group ($p<0.001$), whereas the prevalence of posteromedial cortical comminution was similar between the groups. Overall radiographic parameters, including postoperative displacement, neck–shaft angle, GAI, and femoral neck shortening, did not differ significantly between the groups. Acceptable reduction according to GAI was achieved in 77.8%, and excellent-to-good reduction according to the HRC was achieved in 60%. Assessment using the HRC demonstrated a higher proportion of excellent reductions in the open reduction group (53.3% vs. 16.9%, $p=0.012$).

Serial analysis of the NSA revealed progressive varus collapse in both groups. In the closed reduction group, the mean NSA decreased from 136.2° postoperatively to 132.8° at 3 months and 131.7° at 12 months ($p<0.001$). In the open reduction group, the mean NSA decreased from 136.1° to 127.4° and 129.0° at the corresponding time points ($p=0.001$). Post hoc analyses showed that most varus changes occurred within the first 3 months, with no significant progression thereafter.

Femoral neck shortening at 3 and 12 months was similar between the groups, with no significant differences observed

Table 2 Postoperative radiographic outcomes according to the reduction method

	Total (n=98)	CR group (n=83)	OR group (n=15)	p value
<i>Pauwels classification</i>				<0.001*
1	4	4	0	
2	35	35	0	
3	59	44	15	
<i>Posteromedial cortex comminution</i>				0.79
Yes	43	37	6	
No	55	46	9	
<i>Neck shaft angle</i>				
Contralateral side (Healthy side)	138.5±4.9	138.4±4.7	139.3±6.1	0.60
Post-operation	136.2±6.8	136.2±7.0	136.1±6.0	0.87
3 months follow-up	131.9±9.1	132.8±8.6	127.4±11.0	0.07
12 months follow-up	131.4±8.3	131.7±8.4	129.0±7.5	0.26
Post-operation displacement(mm)	1.8±2.4	2.0±2.5	0.8±1.7	0.08
<i>GAI</i>				0.56
Anatomic	0	0	0	
Acceptable	76	63	13	
Borderline	14	13	1	
Unacceptable	8	7	1	
<i>HRC</i>				0.01*
Excellent	22	14	8	
Good	37	35	2	
Fair	34	29	5	
Poor	5	5	0	
<i>Shortening(mm)</i>				
3 months (mm)	5.9±6.6	5.7±6.0	6.9±9.4	0.89
>5 mm		45	7	0.58
>10 mm		15	5	0.22
12 months (mm)	6.9±6.5	7.0±6.6	6.5±6.8	0.82
>5 mm		43	7	0.97
>10 mm		19	4	0.60

GAI Garden alignment index, HRC Haidukewych's reduction criteria

* Statistically significant

in clinically relevant (> 5 mm) or severe (> 10 mm) shortening. Intraobserver reliability for all radiographic measurements was good (ICC>0.78; κ >0.76; both p <0.001).

Multivariate analysis of complications in the overall cohort

Multivariate analysis of the overall cohort identified age as an independent predictor of non-union (odds ratio [OR]=1.089, p =0.024) (Table 3). Reduction quality assessed by the HRC and posterior cortical comminution were associated with conversion to arthroplasty, with fair

Table 3 Multivariate logistic regression of complications in the overall cohort

Outcome	Predictor	Odds ratio	Estimated 95% CI	p value
<i>Non-union</i>				
	Age	1.089	1.011–1.174	0.024*
	Length of stay	1.134	1.010–1.274	0.033
	Haidukewych's classification			0.017
<i>Excellent(ref)</i>				
	Good	0.379	0.041–3.481	0.391
	Fair	4.13	0.646–26.395	0.134
	Poor	9.449	0.609–146.569	0.108
<i>Osteonecrosis</i>				
	Sex	4.21	1.046–16.951	0.043*
<i>Conversion to arthroplasty</i>				
	Posterior cortex comminution	0.116	0.020–0.686	0.017*
	Haidukewych's classification			0.042*
<i>Excellent(ref)</i>				
	Good	16.578	1.224–224.501	0.035*
	Fair	38.411	3.155–467.592	0.004*
	Poor	11.929	0.089–1591.021	0.321

* Statistically significant

CI confidence interval

and poor reductions conferring substantially higher odds than excellent reductions. Osteonecrosis was associated with patient sex (OR=4.21, p =0.043) but showed no significant association with reduction-related variables.

Multivariate analysis in Pauwels type III fractures

Given the imbalance in fracture severity between the groups, additional multivariate analyses were performed in patients with Pauwels type III fractures (Table 4). Within this high-risk subgroup, non-union was associated with older age, greater intraoperative blood loss, and poorer reduction, after adjustment for confounders. Osteonecrosis was associated with older age (OR=0.935, p =0.037). Reduction quality assessed by the HRC was associated with major reoperation (p =0.032), with fair reduction conferring higher odds than excellent reduction (OR=23.49, p =0.005). Delayed surgery showed a trend towards predicting major reoperation (p =0.05). Femoral neck shortening>10 mm at 3 months was associated with older age (OR=1.132, p =0.038) and open reduction (OR=51.02, p =0.01). At 12 months, older age and intraoperative blood loss remained significant predictors.

Table 4 Multivariate logistic regression of biological and mechanical failure in pauwels type III femoral neck fractures

Outcome	Predictor	Odds ratio	Estimated 95% CI	<i>p</i> value
<i>Biological failure</i>				
<i>Non-union</i>				
	Age	1.104	1.002–1.216	0.045*
	Blood loss	1.004	1.001–1.008	0.025*
	Haidukewych's classification			0.042*
<i>Excellent (ref)</i>				
	Good	0.042	0.001–2.190	0.116
	Fair	9.844	1.592–60.877	0.014*
	Poor	-	-	0.999
<i>Osteonecrosis</i>				
	Age	0.935	0.878–0.996	0.037*
<i>Major reoperation</i>				
	Injury to operation (hours)	1.024	1.000–1.048	0.05
	Haidukewych's classification			0.032*
<i>Excellent (ref)</i>				
	Good	3.293	0.459–23.625	0.236
	Fair	23.488	2.618–210.710	0.005*
	Poor	3.334	0.059–188.561	0.559
<i>Mechanical failure</i>				
<i>Femoral neck shortening > 10 mm (3 mons)</i>				
	Age	1.132	1.007–1.272	0.038*
	Open reduction	51.02	2.602–1000.203	0.01*
	Injury to operation	0.968	0.939–0.997	0.034*
<i>Femoral neck shortening > 10 mm (12mons)</i>				
	Age	1.17	1.052–1.301	0.004*
	Blood loss	1.005	1.001–1.008	0.026*

* Statistically significant

CI confidence interval

Discussion

Preservation of the native femoral head remains the primary goal in non-elderly patients with DFNFs because of the limited longevity and long-term consequences of hip arthroplasty in this population [26]. However, fixation failure remains common, with reported rates of non-union, osteonecrosis, conversion to arthroplasty, and reoperation ranging from 9 to 30%, 14–18%, 14–24%, and 50%, respectively [9–11, 16, 27]. Although closed reduction is often preferred as the initial strategy because of its perceived advantage of preserving femoral head vascularity, open reduction is sometimes required to achieve anatomical alignment in irreducible or highly unstable fractures. In our cohort, the overall outcomes, including non-union, osteonecrosis, reoperation, and conversion to arthroplasty, were comparable to those reported in contemporary series.

Achieving anatomical reduction in DFNFs remains technically demanding, regardless of whether closed or open techniques are used. Soft-tissue interposition, cortical impaction, and unfavorable fracture morphology can prevent successful closed reduction, whereas three-dimensional deformities, particularly residual varus or posterior tilt, are frequently underestimated on fluoroscopic assessment [28, 29]. Previous studies have consistently demonstrated that poor reduction quality is a strong predictor of fixation failure, with reported failure rates approaching 80% when acceptable alignment is not achieved [12]. In our cohort, fractures with relatively stable configurations were generally reduced and stabilized satisfactorily, whereas the most unstable patterns, predominantly Pauwels type III, were overrepresented among patients requiring open reduction. These vertically oriented fractures pose the greatest challenge to achieving and maintaining anatomic reduction and, therefore, warrant focused discussion of fixation strategies in this high-risk subgroup.

Pauwels type III femoral neck fractures are characterized by high vertical shear forces and inherent instability. Accordingly, fixation constructs must provide robust resistance to shear and rotation. Contemporary biomechanical and clinical evidence favors fixed-angle or angular-stable constructs, such as an SHS with anti-rotation screw (s) or the FNS, over multiple CSs in vertical fracture patterns [10, 30, 31]. An MBP has been proposed as an adjunct to enhance stability by neutralizing inferomedial sliding and converting shear forces into compression across the fracture plane, thereby improving the mechanical environment for union [32, 33].

In a large multicenter study of 565 young adults with DFNFs, Roser et al. reported higher failure rates with CSs (56%) than with fixed-angle constructs (36%). Among fixed-angle strategies, SHS combined with an anti-rotation screw and MBP demonstrated the lowest failure rate (11%), supporting the importance of implant selection and construct stability in vertically unstable fractures [10]. In the present cohort, among the open reduction cases with MBP augmentation, cannulated screws were used more frequently than fixed-angle fixation, which may have contributed to the relatively high complication rate observed in the open reduction group. In selected cases, InterTAN was used as an angular-stable cephalocervical construct with integrated rotational control, and previous biomechanical studies have suggested its potential advantages in unstable Pauwels type III fracture models [34].

Importantly, the MBP was used as an adjunctive inferomedial support rather than the primary load-bearing fixation construct. Overall construct stability still primarily depended on the lateral fixation device and fracture healing progression. In our failed cases, the predominant failure

pattern was early varus collapse associated with breakage of the medial plate, suggesting persistent mechanical loading across the construct before fracture union. Therefore, although MBP augmentation may improve medial support, it may not fully compensate for insufficient lateral fixation stability in highly vertical and unstable fracture patterns. Given that the application of MBP generally requires open reduction, particularly in Pauwels type III fractures where indirect reduction is often insufficient, these unstable fracture patterns remained prone to early femoral neck shortening and varus settling during the postoperative period.

In our cohort, the NSA decreased significantly from the immediate postoperative period to 3 months in both the closed and open reduction groups, with no further significant progression. This temporal pattern aligns with prior reports showing that femoral neck collapse, manifesting as shortening and varus deformity, occurs predominantly in the early postoperative phase before plateauing [23, 35]. Chen et al. reported that most shortening occurs within the first 1–3 months after fixation, whereas Polat et al. observed a median onset at 30 days, supporting the concept that clinically relevant collapse represents an early mechanical phenomenon rather than late loss of reduction [23, 35].

Therefore, the early decrease in the NSA observed in our study is more plausibly attributable to the mechanical settling of an intrinsically unstable intracapsular fracture under physiological load. Fracture characteristics such as high Pauwels angles, medial or posteromedial comminution, and residual varus alignment are well-established risk factors for early mechanical compromise and femoral neck shortening [13, 16, 23, 36, 37]. Accordingly, the greater early varus change observed in the open reduction group most likely reflects the preferential selection of more unstable fracture patterns rather than a detrimental effect of the open approach itself. Although concerns have been raised regarding greater soft-tissue disruption with open reduction, current evidence does not demonstrate a higher incidence of non-union or osteonecrosis attributable to open reduction and internal fixation [16, 38]. When performed through an anterior approach with a planned capsulotomy, such as the Smith–Petersen approach, preservation of the ascending cervical arterioles supplying the femoral head is achievable, supporting the concept that osteonecrosis is primarily related to the initial fracture insult rather than to surgical exposure itself [39, 40]. Taken together, our findings suggest that the choice between closed and open reduction should be guided by fracture morphology and the ability to achieve stable anatomical reduction, rather than by the expectation of superior outcomes with either approach alone.

Despite efforts to minimize bias, this study had several limitations. First, the retrospective design carries the inherent risks of selection and information bias, and the small

single-center cohort may limit the generalizability of our findings. Furthermore, the reviewers were not fully blinded to the surgical approach because the implant configuration was visible on the radiographs. Using 60 years as the upper age limit introduced greater heterogeneity in injury mechanisms and fracture characteristics, including a higher proportion of lower-energy injuries in older patients within the cohort. This may have influenced the treatment selection and complication profiles despite the subgroup analysis of Pauwels type III fractures. The modest sample size also limited the statistical power to identify independent risk factors after stratification by surgical approach. Because of the relatively small sample size and unequal group distribution, some analyses required non-parametric statistical methods, which may have reduced statistical sensitivity. In addition, the involvement of multiple orthopedic surgeons may have introduced variability related to differences in surgical technique and decision-making. The decision to perform open or closed reduction was not randomized. Therefore, selection bias is inherent. Although all procedures were performed or supervised by attending orthopedic trauma surgeons, surgeon-related variability in fixation technique and implant configuration could not be fully assessed in this retrospective study and may have influenced the postoperative outcomes. Radiographic assessment is the primary method for evaluating reduction quality and postoperative alignment; however, more precise modalities, such as computed tomography, are not routinely available. Variations in patient positioning during radiographic acquisition may also have affected the angular measurements. Although a standardized postoperative weight-bearing rehabilitation protocol was recommended, actual compliance could not be objectively confirmed in every patient. Therefore, premature weight bearing may have contributed to early mechanical failure in some cases. Future prospective multicenter studies with larger sample sizes and standardized imaging protocols, including advanced cross-sectional imaging, are warranted to better define the outcome determinants and validate our findings.

In conclusion in young patients with DFNFs, fracture morphology and patient age were more influential determinants of outcomes than the choice between closed and open reduction. Open reduction achieved a higher proportion of excellent reductions but did not reduce complication rates compared with closed reduction, highlighting the importance of fracture morphology-guided fixation and realistic patient counseling.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval This retrospective chart review study involving human participants was conducted in accordance with the ethical standards of the institutional and national research committee and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards. The study protocol was approved by the Institutional Review Board (IRB NO: 202600177B0) of Chang Gung Memorial Hospital, Taoyuan City, Taiwan.

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